Data analysis and the representation of data

Collecting numbers is the easy part — this lesson is about turning them into a picture that tells the truth.

LESSON 79–80 | 2 × 40 MIN | ALGEBRA 8, CHAPTER IV §26 | STAGE 9 · 6.1, 15.1, 15.5

LESSON CLOCK

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Collect data and organise it in a frequency table.
- Choose a suitable chart for a given kind of data.
- Draw and read a frequency polygon.
- Recognise a misleading graph.

TEXTBOOK

National	Chapter IV, pp. 165–172
Cambridge	Learner's Book pp. 129–137, 318–330
Workbook	Workbook 6.1, 15.1

01 Explanation

From raw numbers to a table

A list of 40 test marks tells you nothing at a glance. A **frequency table** does.

MARK	0–10	10–20	20–30	30–40	40–50
Frequency	2	5	8	6	3

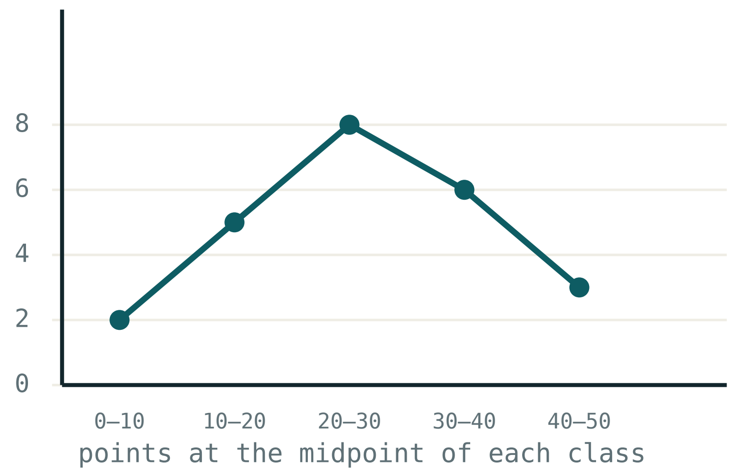
The groups are **classes**. Keep them equal in width, and say clearly which end belongs where — "10–20" usually means $10 \le x < 20$.

SAMPLING

When you cannot ask everybody, you ask a **sample**. A good sample is large enough and chosen at random. Asking only your friends, or only people outside a sports hall, produces **bias** — the answer is wrong before any arithmetic starts.

Which chart?

CHART	USE IT FOR	WATCH OUT FOR
Bar chart	comparing separate categories	bars must be equal width
Pie chart	showing parts of one whole	useless for comparing two different totals
Line graph	change over time	only for continuous data
Frequency polygon	the shape of a grouped distribution	plot at the class midpoint



A frequency polygon: plot each frequency above the midpoint of its class, then join the points with straight lines.

The polygon shows the **shape** of the distribution — where the bulk of the data sits, and whether it leans to one side.

Misleading graphs

THREE TRICKS TO LOOK FOR

1. **A vertical axis that does not start at zero** — small differences look enormous.
2. **Unequal class widths** drawn as equal bars.
3. **Pictures scaled in two directions** — doubling the height and the width multiplies the apparent size by four, not two.

Ask of every graph: what is on each axis, where does the scale start, and does the picture match the numbers?

02 Worked examples

EXAMPLE 1 **Twenty marks are: 5, 12, 18, 23, 25, 27, 29, 31, 33, 34, 36, 38, 41, 44, 8, 15, 22, 26, 30, 35. Make a frequency table with classes of width 10.**

1. $0–10: 5, 8 \to 2$

   Tally each value into its class.

2. $10–20: 12, 18, 15 \to 3$

3. $20–30: 23, 25, 27, 29, 22, 26 \to 6$

4. $30–40: 31, 33, 34, 36, 38, 30, 35 \to 7$

5. $40–50: 41, 44 \to 2$

   Total $2 + 3 + 6 + 7 + 2 = 20$ ✓

ANSWER

$2, 3, 6, 7, 2$

EXAMPLE 2 **Which chart suits "the number of learners choosing each of four sports"?**

1. Four separate categories, not a time series.

2. A bar chart compares categories directly.

3. A pie chart also works if you want the share of the whole class.

   But it hides the actual numbers.

ANSWER

A bar chart (or a pie chart for proportions).

EXAMPLE 3 **A graph shows sales rising from 98 to 102 with the axis starting at 96. Why is it misleading?**

1. The rise is *4* out of *98* — about *4%*.
2. With the axis starting at 96, the bar for 102 looks three times the bar for 98.
3. Redrawing from zero shows the two bars almost equal.

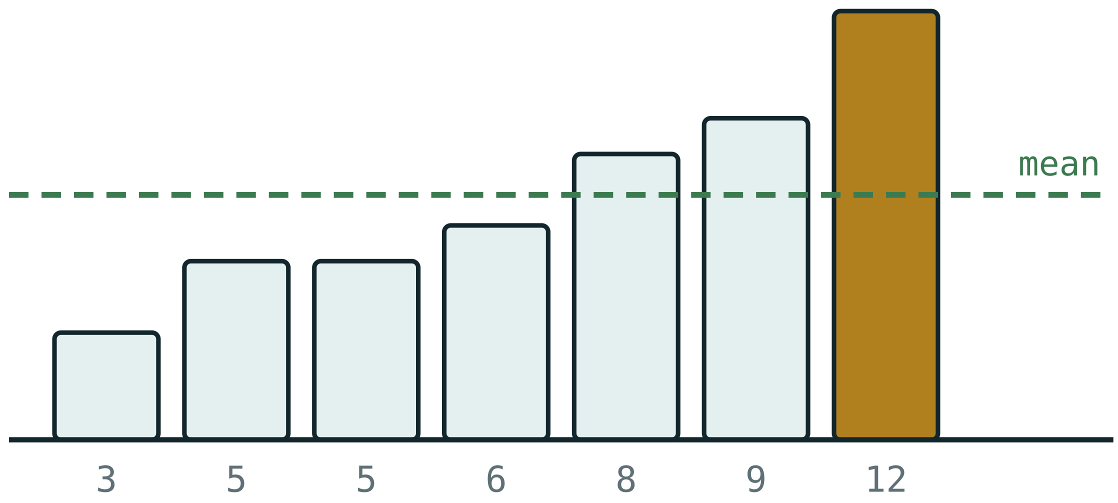
ANSWER

The truncated axis exaggerates a small change.

03 Interactive model

Change one value and watch which average moves — the mean always, the median sometimes, the mode rarely.

What the data looks like



values **3, 5, 5, 6, 8, 9, 12** | mean **6.86** | median **6** | mode **5** | range **9**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Data	Ma’lumotlar	Данные
Frequency	Chastota	Частота
Frequency table	Chastotalar jadvali	Таблица частот
Class (interval)	Sinf (oraliq)	Класс (интервал)
Bar chart	Ustunli diagramma	Столбчатая диаграмма
Pie chart	Doiraviy diagramma	Круговая диаграмма
Frequency polygon	Chastotalar ko‘pburchagi	Полигон частот
Sample	Tanlanma	Выборка
Bias	Nomutanosiblik	Смещение (необъективность)
Misleading graph	Chalg‘ituvchi grafik	Вводящий в заблуждение график

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A frequency polygon is plotted at:

- the class boundaries
- the class midpoints
- the largest value
- zero

A pie chart is best for:

- change over time
- parts of one whole
- comparing two different totals
- raw data

Asking only your friends creates:

- a large sample
- bias
- a frequency table
- a fair sample

A vertical axis starting at 96 instead of 0:

- is always wrong
- exaggerates small differences
- hides large differences
- makes no difference

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *What is the frequency of a value?*

   ANSWER How many times it occurs.

2. *Marks 2, 5, 5, 7, 5. Find the frequency of 5.*

   ANSWER *3*

3. *Which chart shows parts of a whole?*

   ANSWER A pie chart.

4. *Which chart shows change over time?*

   ANSWER A line graph.

5. *A class has 8, 12, 15, 5 learners in four groups. Find the total.*

   ANSWER *40*

6. *In a frequency polygon, where is each point plotted?*

   ANSWER Above the midpoint of the class.

7. *Class 20–30. Find its midpoint.*

   ANSWER *25*

Medium · the standard question

7 PROBLEMS

1. *Marks 5, 12, 18, 23, 25, 27, 29, 31. Make classes of width 10 and give the frequencies.*

   ANSWER *0–10: 1, 10–20: 2, 20–30: 4, 30–40: 1*

2. *Frequencies 3, 7, 10, 6, 4. Find the total and the modal class.*

   ANSWER *30*; the third class

3. *Which chart for "learners choosing each of four sports"?*

   ANSWER A bar chart.

4. *A pie chart has 360°. What angle represents 25%?*

   ANSWER *90°*

5. *A pie chart has 40 learners. What angle represents 10 of them?*

   ANSWER *90°*

6. *Why is asking people outside a gym about exercise biased?*

   ANSWER They are far more likely to exercise than the general population.

7. *Class midpoints 5, 15, 25, 35 with frequencies 2, 5, 8, 3. Which class is modal?*

   ANSWER *20–30*

Hard · stretch and reason

7 PROBLEMS

1. *A graph shows sales rising 98 → 102 with the axis from 96. Explain why it misleads.*

   ANSWER A 4% rise is drawn to look like a threefold one; the axis should start at 0.

2. *A pie chart of 90 learners: one sector is 120°. How many learners is that?*

   ANSWER $\frac{120}{360} \times 90 = 30$

3. *Frequencies 4, 9, 12, 7, 3 for classes of width 5 starting at 0. Draw the frequency polygon points.*

   ANSWER $(2.5; 4), (7.5; 9), (12.5; 12), (17.5; 7), (22.5; 3)$

4. *A survey of 20 people finds 15 like a product. Is that enough to claim 75% of the town does?*

   ANSWER No — the sample is small and may not be random; the margin of uncertainty is large.

5. *Two pie charts show 60% for one school of 100 and 60% for another of 1000. Are the sectors comparable?*

   ANSWER The angles are equal, but they represent 60 and 600 people — a pie chart hides the totals.

6. *Explain why doubling both the width and the height of a picture symbol misleads.*

   ANSWER The area becomes four times as large, so the eye reads a fourfold increase.

7. *Design a fair way to sample 50 learners from a school of 800.*

   ANSWER Number everyone 1–800 and choose 50 at random, or take a proportional number from each year group.

07 Homework

Homework — 5 problems

Algebra 8, Chapter IV, pp. 165–172, and Cambridge Learner's Book 6.1.

1. *Marks 3, 9, 14, 16, 21, 24, 25, 28, 31, 37. Make a frequency table with classes of width 10.*
2. *Draw the frequency polygon for that table.*

3. *A pie chart of 72 learners has a sector of 150°. How many learners is that?*
4. Name a suitable chart for each: monthly temperature; favourite subject; share of a budget.
5. Describe one way a graph in a newspaper could mislead, and how to check for it.